AMAN YADAV

amandkyadav@gmail.com | 9428587817 | linkedin.com/in/amandkyadav | tryhackme/Automatlog

Full-stack engineer with a security backbone — ships AI-native features (Claude Code, RAG, MCP) and hardens them. Built unified OAuth across 6+ platforms and multi-channel CPaaS pipelines, and found critical vulnerabilities (SQLi, IDOR, auth bypass) in a production gateway via a self-led OWASP assessment. Independent MCP / agentic-security research. Azure- & Google Cloud–certified; ISO/IEC 27001.

EXPERIENCE

Smart Technology — *Software Developer* *Sept 2025 – Present*

Full-stack + platform engineering across CPaaS, AI, and security, in a forward-deployed delivery model.

- Built unified OAuth hubs managing consent and token lifecycles across 6+ platforms (Zoho, HubSpot, Shopify, Razorpay), consolidating six separate flows into one auditable architecture.
- Engineered multi-channel pipelines (WhatsApp, SMS, RCS) integrating major CRMs and payment gateways with webhook signature validation and access-scoping.
- Managed API integrations across Google Cloud services (Sheets API, Workspace Marketplace) and undocumented/legacy endpoints, including AI features such as OCR order intake — evaluating tool-permission and data risk while maintaining security posture.
- Built cross-platform mobile apps (React Native, Expo) with Firebase, extending core features to iOS and Android.
- Built a RAG pipeline for an AI agent using Python (PyTorch, HuggingFace Transformers), vLLM for model serving, pgvector for vector search, and LangGraph for multi-step orchestration; environments managed with uv/venv.
- Identified critical vulnerabilities — blind SQLi, IDOR, auth bypass, insecure file upload — in a production SMS/messaging gateway via an OWASP-mapped assessment.
- Developed AI-agent tooling, connectors, an npm package, and a real-time voice widget, with multi-model routing across multiple AI-provider endpoints; applied MCP threat-modeling and tool-permission analysis to agent tool execution.
- Implemented WhatsApp Calling API using WebRTC signaling over FreeSWITCH for real-time voice call handling.

Havok Security — *Cybersecurity Internship* *Jul 2024 – Jan 2025*

- Analysed web application vulnerabilities using PortSwigger Web Security Academy tooling; conducted network mapping and traffic analysis for threat detection.
- Completed hands-on SOC operations training through TryHackMe labs, applying Kali Linux tooling for ethical hacking and security assessments.

Hacktify Cybersecurity LLP — *Cyber Security Intern* *Dec 2023 – Jul 2024*

- Configured Linux servers for local deployments and secure remote access.
- Managed Linux environments and automated remote command execution with Ansible.
- Implemented Docker to build containerized environments and intentionally vulnerable machines for security testing.
- Practiced Flaw Cloud labs to master command execution and solve Capture the Flag (CTF) challenges.
- Delivered technical support for WordPress and integrated the CMS with CRM platforms.
- Monitored WordPress vulnerabilities and hardened site security using the Wordfence plugin.
- Performed quality assurance and bug testing to ensure system stability.
- Wrote Python scripts to support penetration testing workflows.

PROJECTS

AI Automation Hub (Automate) *2026*

AI-agent automation platform — visual chatbot builder, multi-agent orchestration with a routing layer, management console, embeddable widgets, a human-like face agent, and a self-learning agent that adapts by following interaction patterns.

AI Agent & MCP Security Research *2026*

Independent threat-modeling research on Model Context Protocol (MCP) servers used in agentic AI systems — tool poisoning, full-schema poisoning, confused-deputy attacks, and prompt/tool misuse — mapped against known CVEs and MCP specification guidance.

Omni CPaaS *2026*

Omnichannel CPaaS solution delivered on Android (React Native, Expo) and iOS, with webhook-driven WhatsApp / SMS / RCS messaging and secure token handling.

GetKhojo (OSINT Security Suite) *2026*

Built a full-stack OSINT scanning platform (React/Python) with an automated PDF reporting pipeline and hardened credential handling (environment-based secrets, no plaintext storage) throughout.

Protein-Ligand Docking Pipeline *2026*

Built an automated molecular docking pipeline (AutoDock-GPU / AutoDock4, OpenBabel, GNINA, RDKit) with AI-assisted result detection and analysis, automating protein-ligand preparation, binding-site docking, and scoring for computational drug-discovery research.

SKILLS

Application Security: OWASP Top 10, Burp Suite, vulnerability assessment, penetration testing (SQLi, IDOR, auth bypass, file-upload vulnerabilities), secure code review

Identity & Access: OAuth 2.0 architecture, multi-platform token/consent management, IAM fundamentals

AI & Agent Security: MCP threat modeling, prompt injection, tool/agent permission risk, agentic system architecture

SOC & Monitoring: Microsoft Sentinel SIEM (lab-based), log analysis, threat detection, incident triage

Cloud & Infrastructure: Azure, AWS (EC2, S3), Google Cloud, Linux hardening, Docker

Compliance: Working familiarity with SOC 2, ISO/IEC 27001, and GDPR principles

Engineering: Python, JavaScript/Node.js, Shell scripting, SQL, CI/CD (Jenkins, GitHub)

EDUCATION

ITM SLS Baroda University *2020 – 2024*

Bachelor of Technology, Computer Science & Engineering — Specialization in Cybersecurity & Networking
Focus on cybersecurity tooling and Capture the Flag (CTF) challenges; GDSC Core Team, Google Cloud Study Jam participant.

COMMUNITY

GDSC (ITMBU) - AWS User Group Vadodara - DeepLearning.AI - Management Lead & Student Ambassador

CERTIFICATIONS

- Microsoft Certified: Azure Administrator Associate (AZ-104) - Sep 2024 - Credential ID 862B527704C9AF53
- ISO/IEC 27001 Information Security Associate (SkillFront) - Sep 2023 - Credential ID 78282553463382
- ISEA Certified Cyber Hygiene Practitioner (MeitY) - Jun 2023 - Credential ID MeitY/ISEA/WCHP/030049
- Foundations of Operationalizing MITRE ATT&CK (AttackIQ)
- Threat Hunting Bootcamp (VTF)
- Mastering AWS Security Strategies and Tactics (VTF)
- Google Cloud Computing Foundations
- TryHackMe - Security Engineer Learning Path (THM-WSLWBW9LUG) - Sep 2023
- SAP Certified Application Associate - SAP Analytics Cloud - Dec 2022 (exp. Dec 2027) - Credential ID iutCVnPq

CURRENT DEVELOPMENT

GenAI evaluation: RAG metrics, model evaluations, LLM-as-a-judge, golden-dataset design, test planning, and safety evaluation (prompt injection, PII, over-refusal).
Building toward: Python evaluation harnesses with RAGAS / DeepEval, trace-based agent and tool-call evaluation, and CI regression gates.